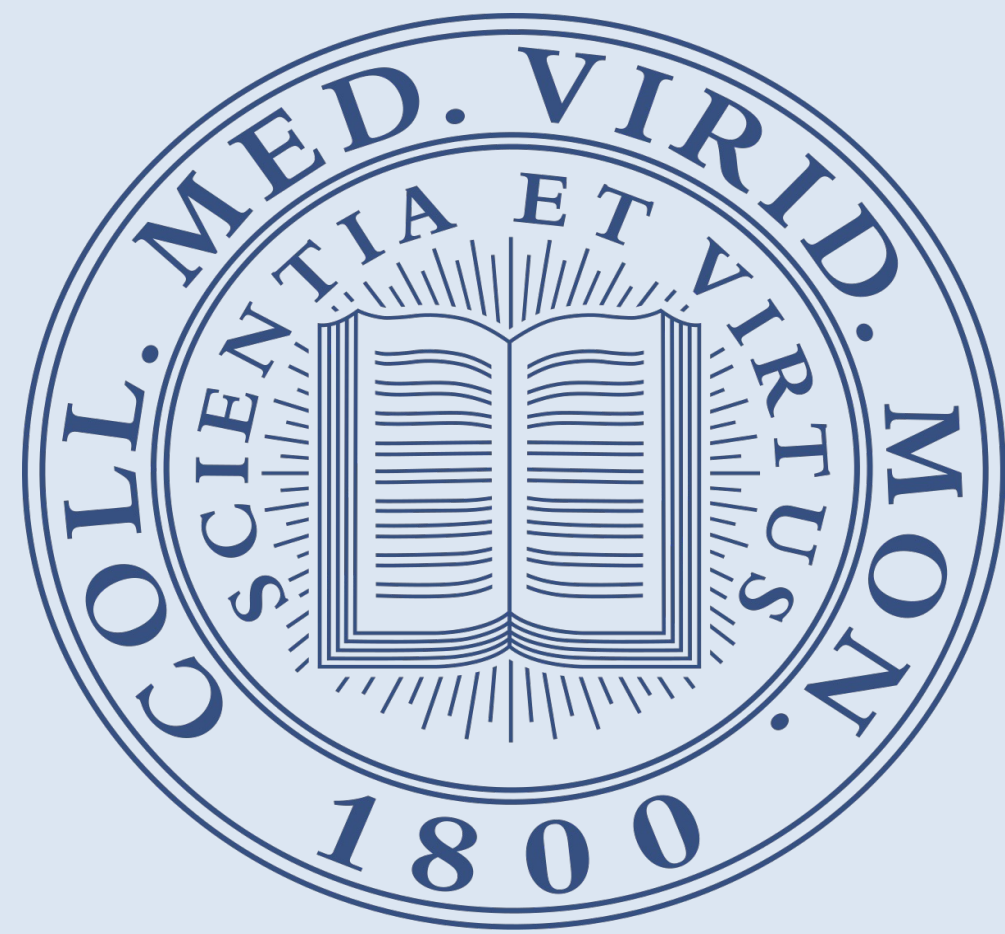




Photocatalytic Demethylation of Dimethylbenzylamine in Oxygen-Free Conditions

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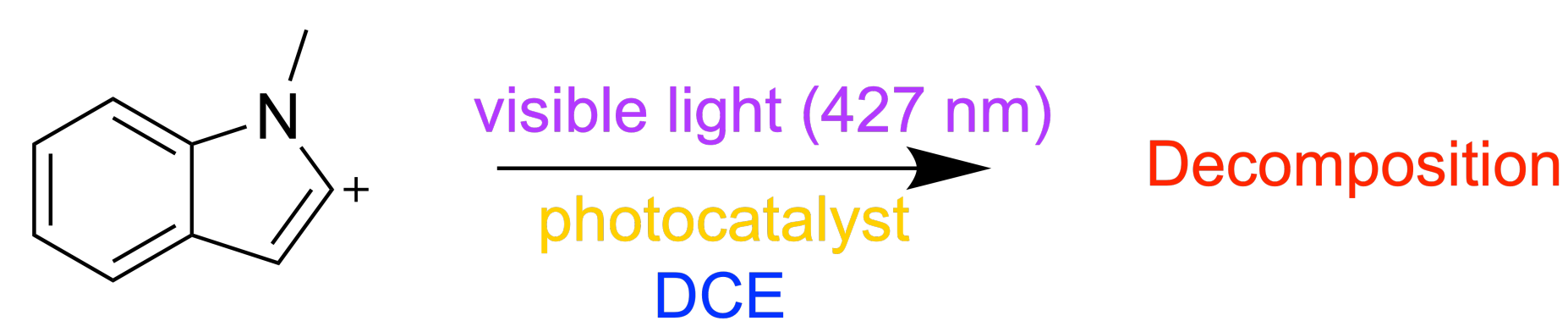
INTRODUCTION

N-demethylation, the removal of an amine methyl group, is an essential transformation in the pharmaceutical world. Drugs are often synthesized from natural products that contain methyl groups. Removing this group opens a site to add functional groups that improve drug efficiency. A common method for selective N-demethylation is the von Braun reaction, which utilizes heat, leading to potential substrate degradation. To minimize degradation, we used photocatalysis, an accessible and mild alternative that uses light to drive the reaction.

HYPOTHESIS

Prior research in the Repka lab showed that MesAcrMe⁺, an acridinium, can demethylate itself when exposed to 427 nm light in dichloroethane (DCE).² From that result, we anticipated that a different acridinium, tBu₂MesAcrPh, that contains a phenyl group instead of a methyl group, could demethylate a substrate. However, initial experiments with vinylic amines resulted in decomposition. We hypothesized that changing the substrate to a benzylic amine, specifically dimethylbenzylamine (DMB), would be a good candidate for photocatalytic N-demethylation, as the amine will not be impacted by resonance.

Previous Work:



Current Hypothesis:

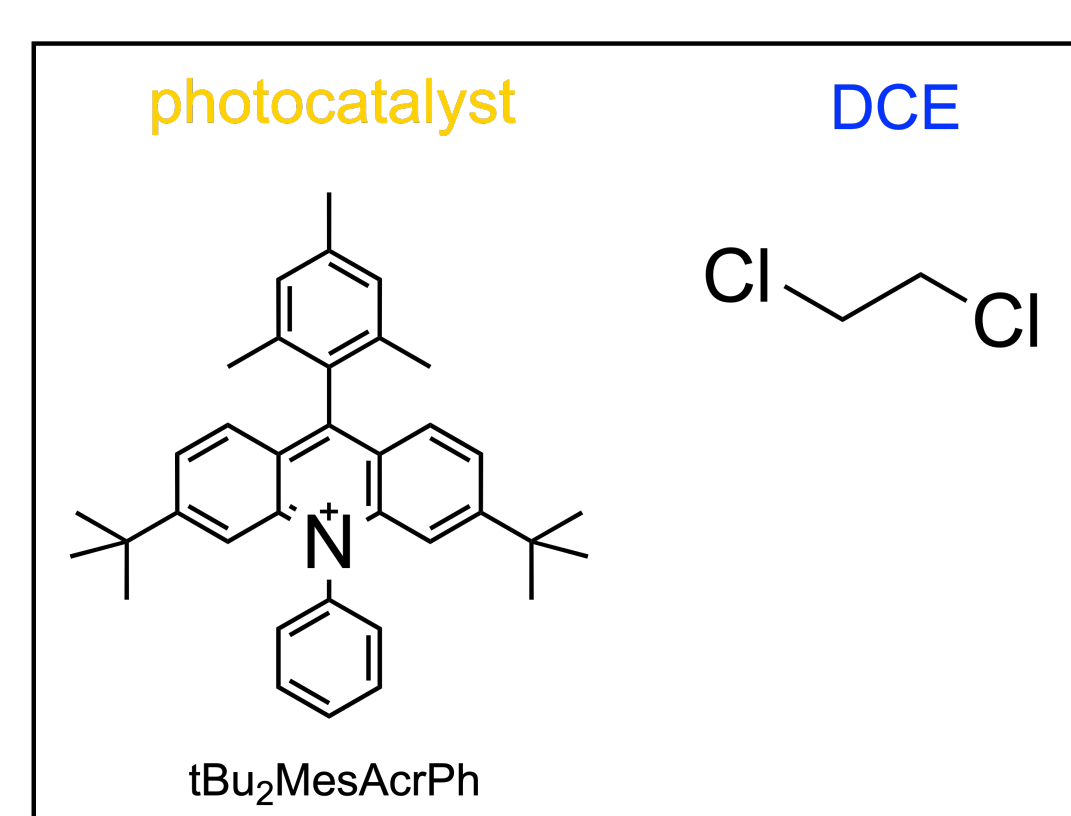
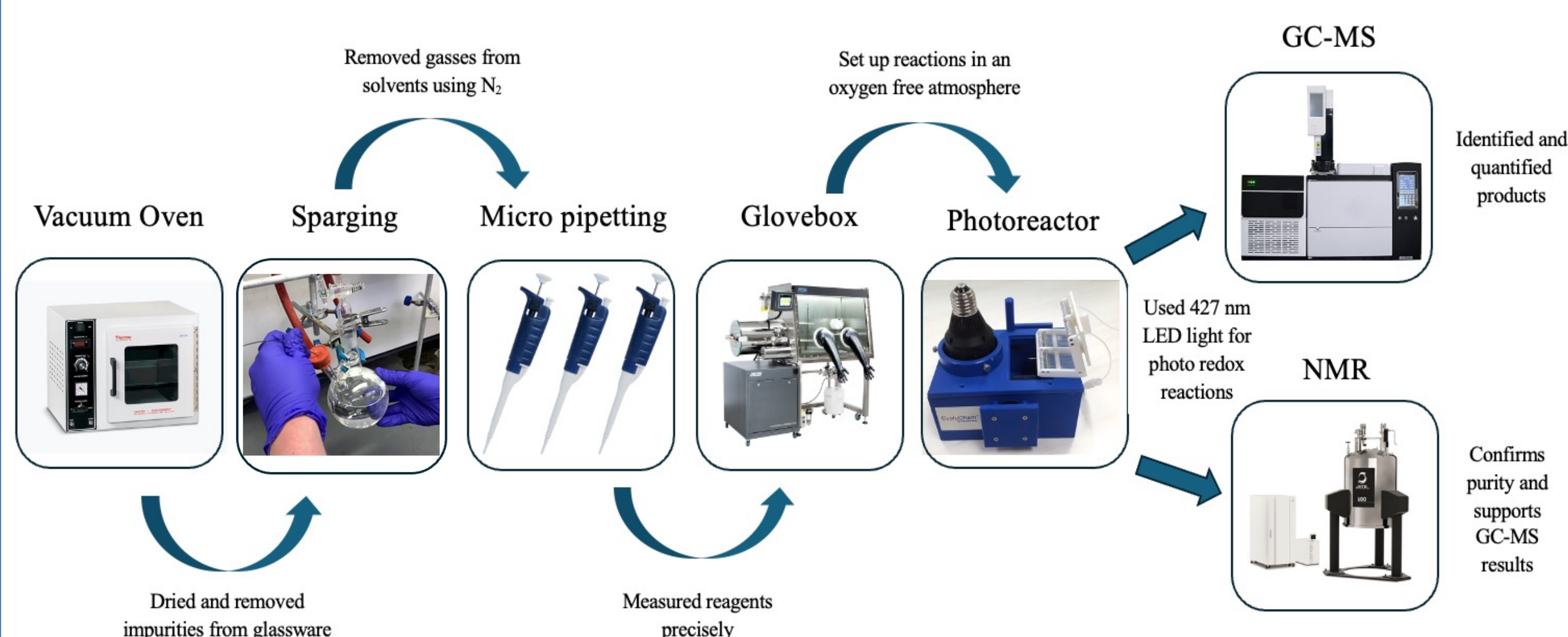


Figure 1. Reaction scheme of the previous work done in the Repka lab showing the decomposition of N-methyl indole using visible light, tBu₂MesAcrPh photocatalyst, and Dichloroethane (DCE). Hypothesized reaction scheme of the demethylation of dimethylbenzylamine to N-methylbenzylamine under the same conditions.

METHODS



RESULTS

Exposure to an acridinium photocatalyst and dichloroethane (DCE) afforded clean demethylation as detected by GCMS and NMR. However, the reaction was non-catalytic as seen by a noticeably low yield and the reaction also produced benzaldehyde. Integrating the GCMS shows that the substrate, dimethylbenzylamine, was ~16 times more prevalent than the product indicating a yield of ~6%. This yield was lower than expected because catalysis wasn't achieved.

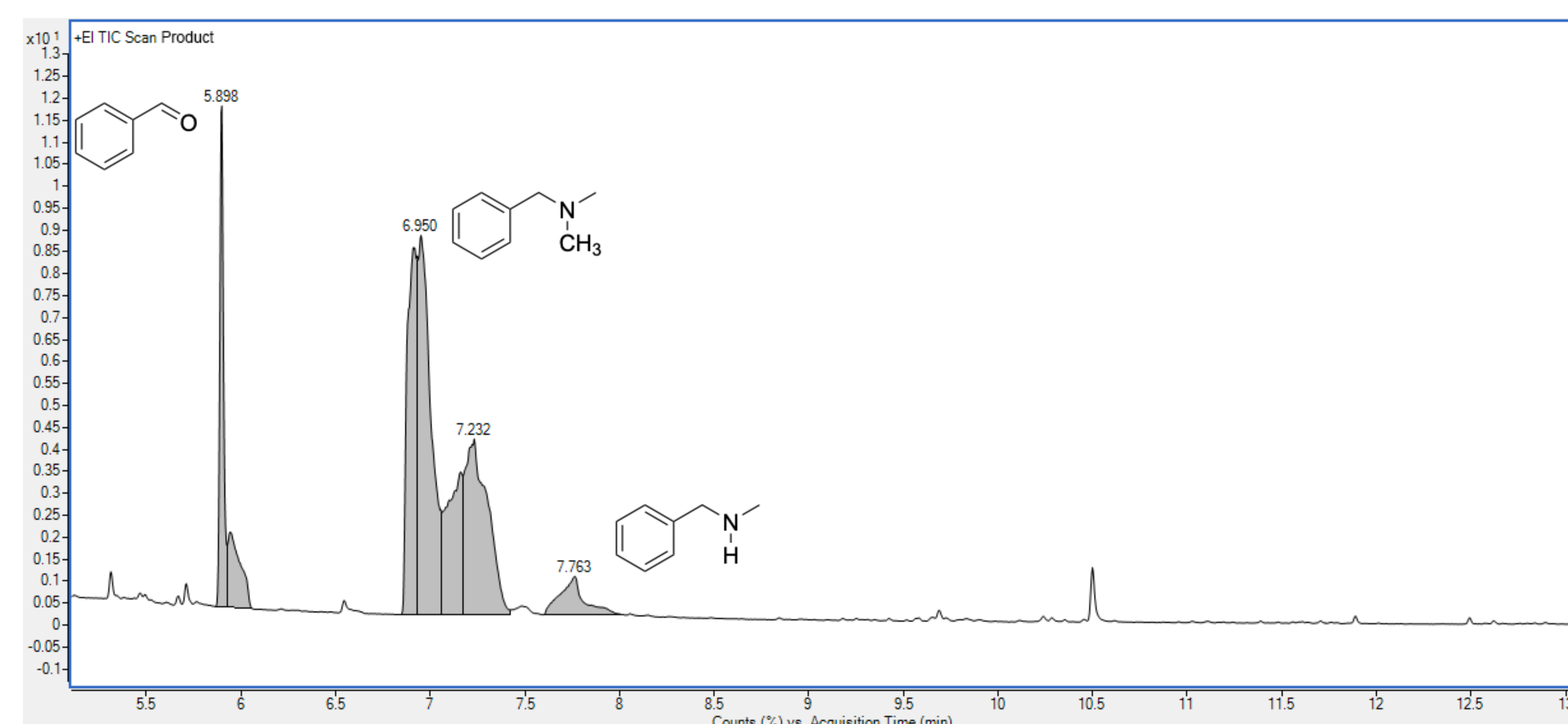
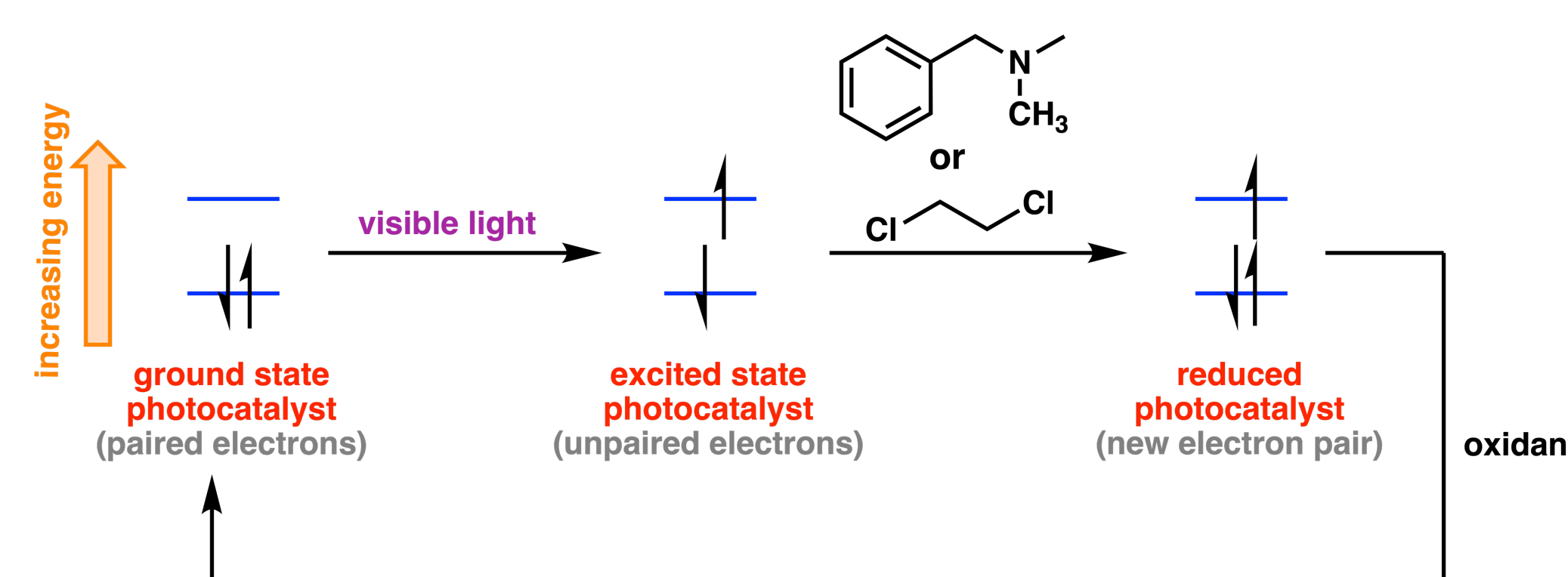


Figure 2. GC-MS analysis of the initial reaction. The integration of the N-methylbenzylamine is small compared to the integration of the dimethylbenzylamine, suggesting a lack of catalytic turnover.

As a result, we re-ran the reaction and incorporated an oxidant to promote catalytic turnover, with the goal of improving both yield and efficiency. The oxidant functions to regenerate the photocatalyst, allowing the reaction to continue beyond a single use of the catalyst.⁴



When the photocatalyst (tBu₂MesAcrPh) absorbs light, an electron is excited, and the catalyst accepts another electron from a substrate. For the reaction to keep going, the catalyst needs to be reset to its original form. Oxidants like ammonium persulfate and diphenyl disulfide promote this by accepting an electron from the reduced catalyst.

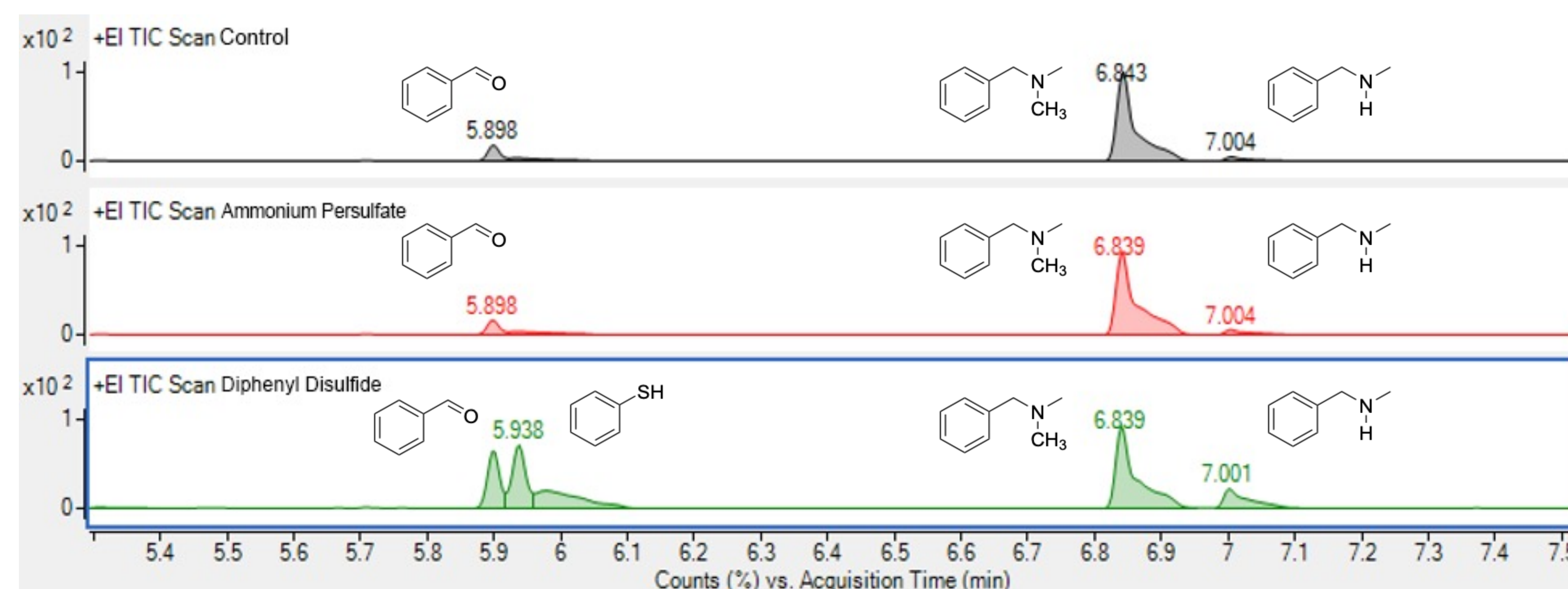


Figure 3. GC-MS analysis of the products, with and without oxidants. The control (top scan) shows small amounts of the predicted product: N-methylbenzylamine. The oxidants, ammonium persulfate (middle scan) and diphenyl disulfide (bottom scan), show enhanced desired product peaks, as well as benzaldehyde, indicating successful catalytic turnover.

FUTURE WORK

Overall, adding an oxidant improved yield and enabled catalytic turnover, but also led to significant benzaldehyde formation due to competing reactions. To address this, we can change the conditions of the experiment by using a different oxidant or changing the solvent, as the reaction is solvent dependent.

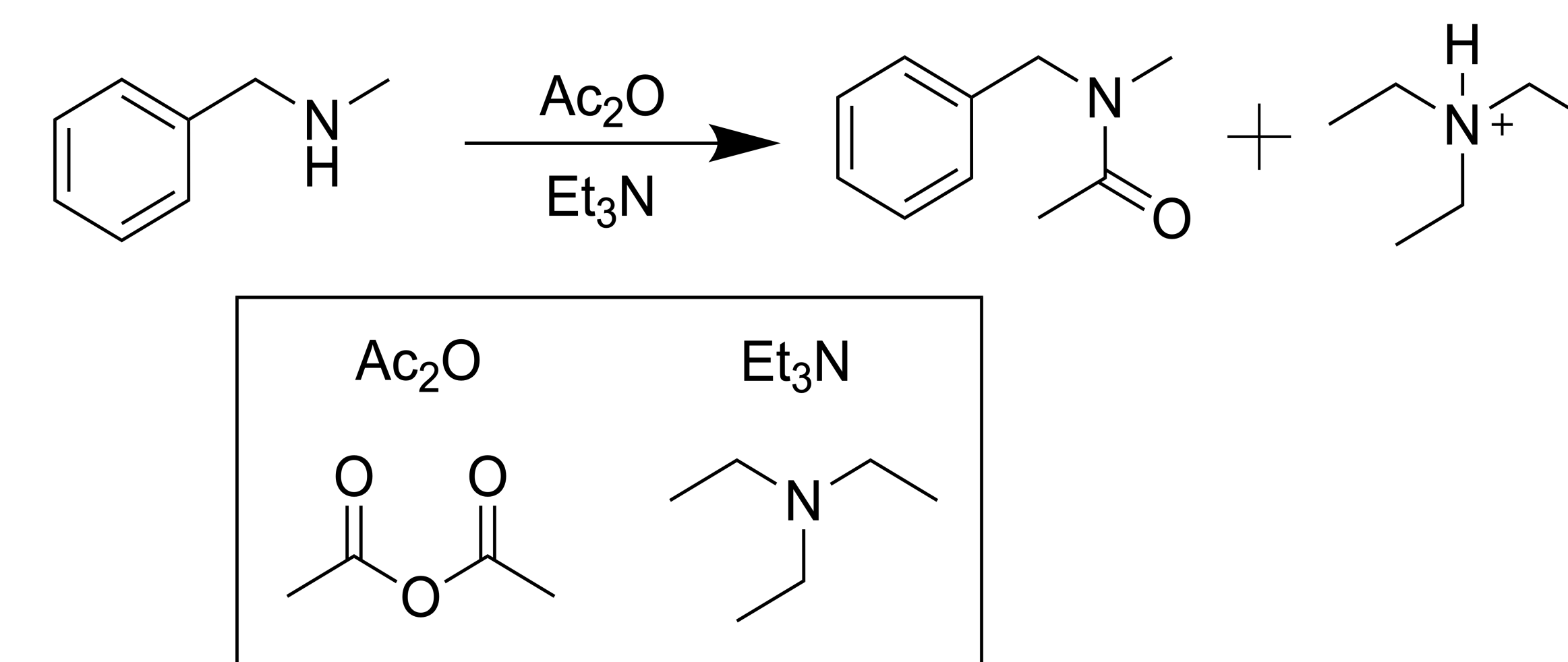


Figure 4. Reaction scheme of the acetylation of the N-methylbenzylamine product.

Acetylation is also an important future step in this research. Purification of the amine product is challenging, and acetylation can make this task straightforward. GC-MS data from our initial acetylation reaction suggests we can use this approach to streamline purification of the product.

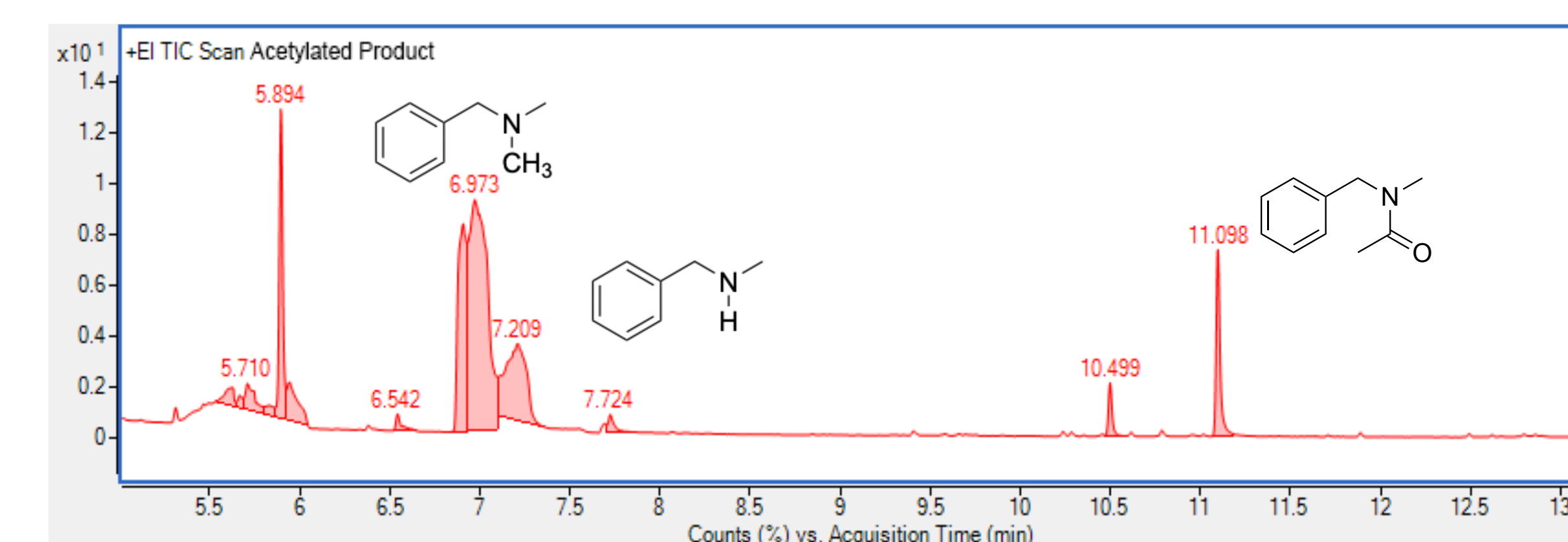


Figure 5. GC-MS analysis showing a successful acetylation of the N-methylbenzylamine product.

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